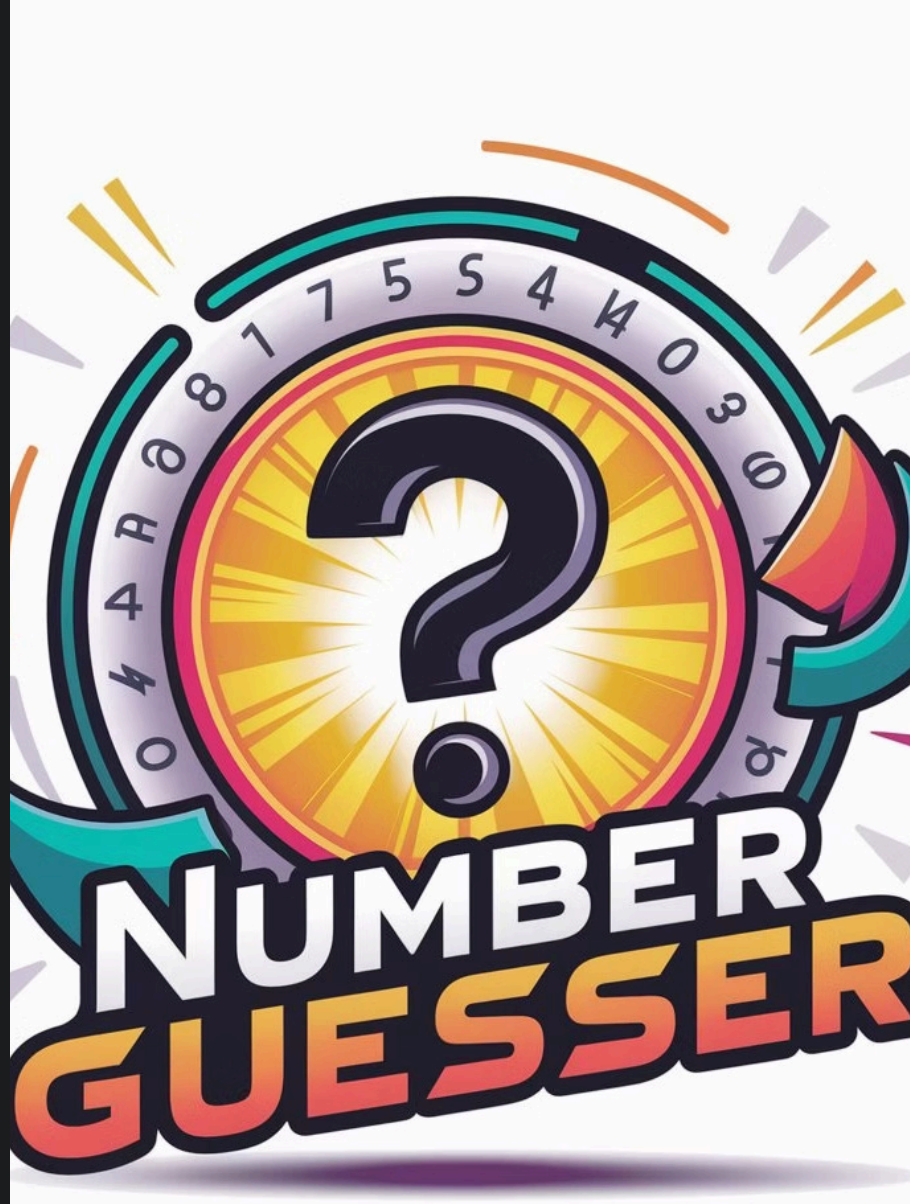
Star Guess Pro 🚀

A Modern Desktop Number Guessing Game



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Team 2

Data Scientists

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Project Concept

What is Star Guess Pro?

Star Guess Pro is an interactive, multi-level desktop game that challenges users to find a secret number within a specific range and limited attempts. Built with Python, the application combines engaging gameplay with a polished visual interface.

The game adapts to different skill levels, providing an appropriate challenge whether you're a casual player or a seasoned number cruncher.

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How It Works

01

Choose Your Difficulty

Select from Easy, Intermediate, or Hard mode to match your skill level

02

Secret Number Generation

The system generates a random secret number based on your chosen level's range

03

Make Your Guess

Enter your guess and receive instant feedback whether you're too high or too low

04

Track Your Progress

Monitor attempts remaining with visual progress indicators and attempt counters

05

Win or Try Again

Discover if you've cracked the code or if you need another attempt

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Technical Stack

Built with Python and designed for Windows, Star Guess Pro leverages lightweight libraries to deliver responsive gameplay without bloated dependencies.

Python

Core programming language providing the game logic and control flow

Tkinter

GUI framework handling windows, frames, and user interactions

Random

Logic module generating unpredictable secret numbers

Messagebox

UI notifications delivering win and game-over alerts

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Library Deep Dive

Tkinter: The Interface Engine

Tkinter creates the modern graphical user interface that brings Star Guess Pro to life. It handles window management, frame organization, and event-driven interactions like button clicks and keyboard input. The framework provides a native look and feel while maintaining cross-platform compatibility.

Random: The Brain of the Game

The Random library generates cryptographic-style secret numbers that ensure a unique experience every time you play. It creates unpredictable values within specified ranges based on difficulty level, making each game session fresh and challenging. The algorithm guarantees fair distribution across the number spectrum.

Messagebox: The Communicator

Integrated throughout the application, Messagebox provides professional pop-up alerts for critical game moments. When players win, lose, or encounter errors, these dialogs deliver clear, actionable feedback. The notifications enhance user experience by confirming game state changes with visual and audio cues.

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Features & Highlights

Multi-Level Difficulty

Three distinct difficulty levels challenge players of all skill levels

Visual Progress Tracking

Progress bars and attempt counters keep players informed

Real-Time Feedback

Instant Higher ⬆️ and Lower ⬇️ hints guide your guesses

Mystery Box Design

Engaging visual elements enhance the gaming experience

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Key Takeaways

Modern Python Desktop Application

Built with industry-standard libraries delivering smooth, responsive gameplay

Accessible to All Skill Levels

Multiple difficulty modes ensure engaging experience whether you're learning or mastering

Polished Visual Experience

Professional interface design with intuitive controls and clear feedback mechanisms

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Thank You

Ready to Play?

Star Guess Pro combines elegant design with engaging gameplay to create an experience you'll want to return to again and again.

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